

Python Programming

From Fundamentals to Real-World Automation

HARZ Digital Learning Series

By Rabiul Hamza Mohammed

Python Programming

HARZ Digital Learning Series

Book: Python Programming

Subtitle: From Fundamentals to Real-World Automation

Author: Rabiul Hamza Mohammed

Publisher: HARZ Digital Services

Copyright 2026 Rabiul Hamza Mohammed. All rights reserved.

For inquiries: hamzarabiu390@gmail.com

Website: <https://rabiulhamza11.github.io/harz-portfolio>

First Edition, 2026

Table of Contents

Chapter 1: Introduction to Python

Chapter 2: Variables, Data Types, and Operators

Chapter 3: Control Flow: If/Else, Loops

Chapter 4: Functions and Modules

Chapter 5: Working with Data Structures

Chapter 6: File Handling and I/O

Chapter 7: Error Handling and Exceptions

Chapter 8: Introduction to Automation

Chapter 9: Web Scraping with Python

Chapter 10: Building Real-World Projects

Chapter 1

Introduction to Python

In this chapter, we will explore introduction to python and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to introduction to python
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to introduction to python. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 2

Variables, Data Types, and Operators

In this chapter, we will explore variables, data types, and operators and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to variables, data types, and operators
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to variables, data types, and operators. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 3

Control Flow: If/Else, Loops

In this chapter, we will explore control flow: if/else, loops and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to control flow: if/else, loops
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to control flow: if/else, loops. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 4

Functions and Modules

In this chapter, we will explore functions and modules and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to functions and modules
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to functions and modules. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 5

Working with Data Structures

In this chapter, we will explore working with data structures and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to working with data structures
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to working with data structures. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 6

File Handling and I/O

In this chapter, we will explore file handling and i/o and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to file handling and i/o
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to file handling and i/o. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 7

Error Handling and Exceptions

In this chapter, we will explore error handling and exceptions and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to error handling and exceptions
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to error handling and exceptions. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 8

Introduction to Automation

In this chapter, we will explore introduction to automation and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to introduction to automation
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to introduction to automation. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 9

Web Scraping with Python

In this chapter, we will explore web scraping with python and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to web scraping with python
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to web scraping with python. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Chapter 10

Building Real-World Projects

In this chapter, we will explore building real-world projects and understand its practical applications in real-world scenarios.

This chapter is designed to give you both theoretical knowledge and hands-on practical skills. Whether you are a beginner or have some experience, the concepts covered here will build a solid foundation for your journey.

Key Topics Covered:

1. Understanding the fundamentals and core concepts
2. Practical applications and real-world examples
3. Step-by-step guides and tutorials
4. Common mistakes and how to avoid them
5. Best practices from industry experts
6. Exercises and practice activities
7. Resources for further learning

Learning Objectives:

By the end of this chapter, you will be able to:

- Explain the key concepts related to building real-world projects
- Apply practical techniques in real situations
- Identify common challenges and solutions
- Build confidence in your skills and knowledge

Summary:

This chapter provides a comprehensive introduction to building real-world projects. The concepts and skills you learn here are essential building blocks for the rest of the book. Take time to practice and review the material before moving on to the next chapter.

Next Steps:

In the next chapter, we will build on these foundations and explore more advanced topics. Make sure you are comfortable with the material in this chapter before proceeding.

Congratulations!

You have completed this book from the HARZ Digital Learning Series.

We hope this book has given you valuable knowledge and practical skills. Continue your learning journey with other books in the series.

Visit: <https://hamzarabiu.gumroad.com>

Contact: hamzarabiu390@gmail.com

Thank you for learning with HARZ Digital Services.